Visualisation II: Scatter Plots and Correlation Analysis

Social and Cultural Analytics (7AAVBCS5)
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KING'S College LONDON

Lecture Plan

- **Part I: Charts as arguments**
- **Part II: Line plots**
- **Part III: Scatter plots**
- **Part IV: Correlations**
- **Part V: Drawing scatter plots that work**
- **Part VI: Thinking about causes** *(extra if time)*
- **Part VII: Workshop**

Key Learning Outcomes

By the end of this week you will be able to:

- Read someone else's chart as an argument, and check it
- Draw a line plot, and say when a line is the wrong chart
- Explore the relationship between two numeric variables
- Read a scatter plot and describe the pattern
- Calculate and interpret Pearson and Spearman correlations
- Choose between them, and justify the choice
- Encode a third variable with size or colour
- Explain why correlation is not causation
- Save a labelled figure in a format fit for a report

The final three teaching weeks

Calendar week	What we will investigate
8 · 23 November	Text analysis, preprocessing, and an introduction to sentiment analysis
9 · 30 November	Topic modelling and an introduction to GIS
10 · 7 December	GIS, social network analysis, and the final assessment

We will learn **loops and conditions through these analytical tasks**.

Read before next week (23 November)

Hill and Nulty, “Contextual Analytics: Reconstructing Meaning in Computational Cultural Analysis”

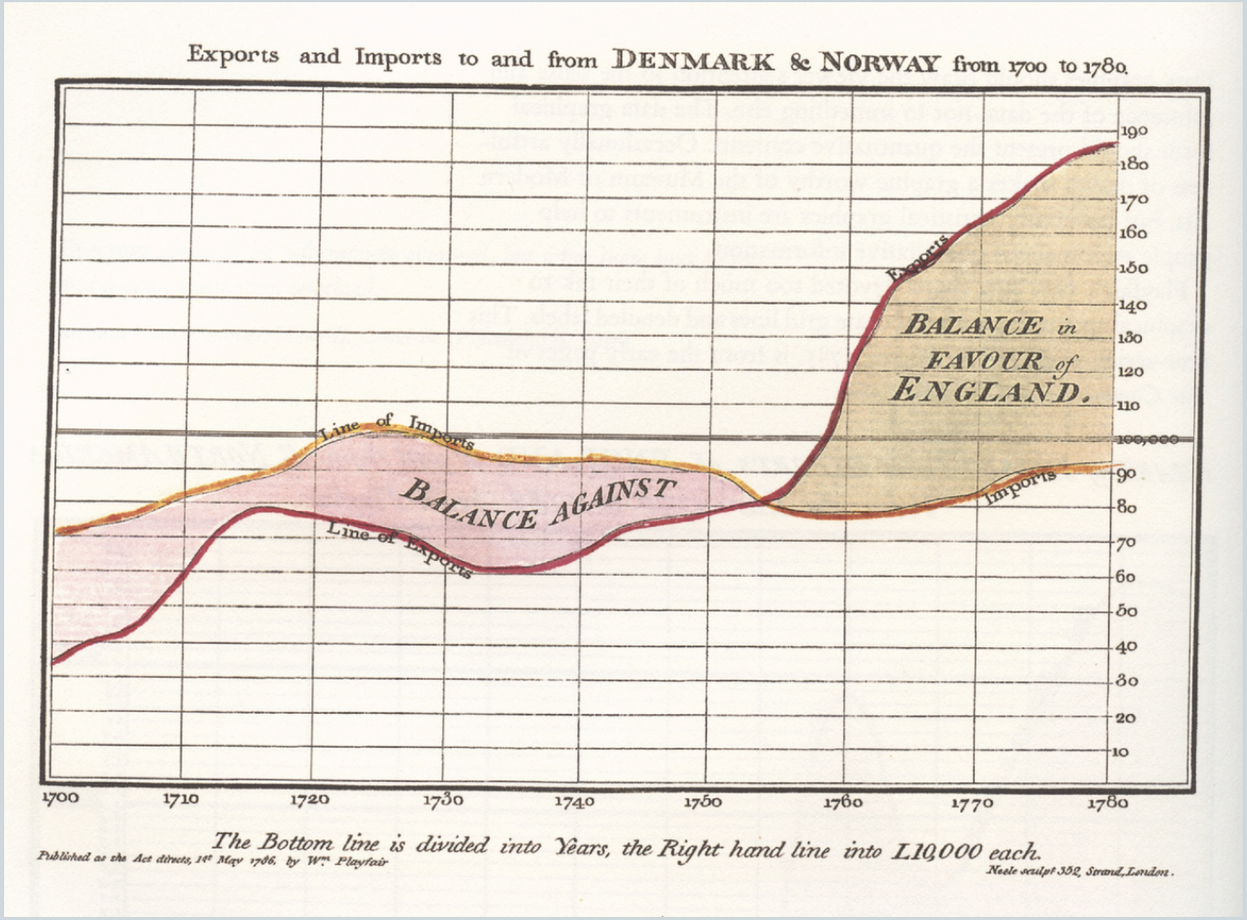
Read the chapter before our next lecture. Pay particular attention to **§3** (five dimensions), **§4** (their interactions) and **§5** (research design).

Think about how these apply to you data. Specifically:

1. Which dimensions does your dataset let you investigate?
2. What changes when an object becomes tokens, a score, coordinates, or edges?
3. What other evidence would you need to support your interpretation?

Part I: Charts as arguments

Playfair: trade as a visual argument



William Playfair, *Commercial and Political Atlas* (1786).

Playfair was a Scottish engineer and political economist, considered the founder of visualising statistics and inventing the line, area, bar, pie, and circle graph. (Was also likely a secret agent for the British Government).

- Years: 1700–1780
- Trade value in pounds; grid divisions £10,000
- Shading: balance “against” or “in favour of England”

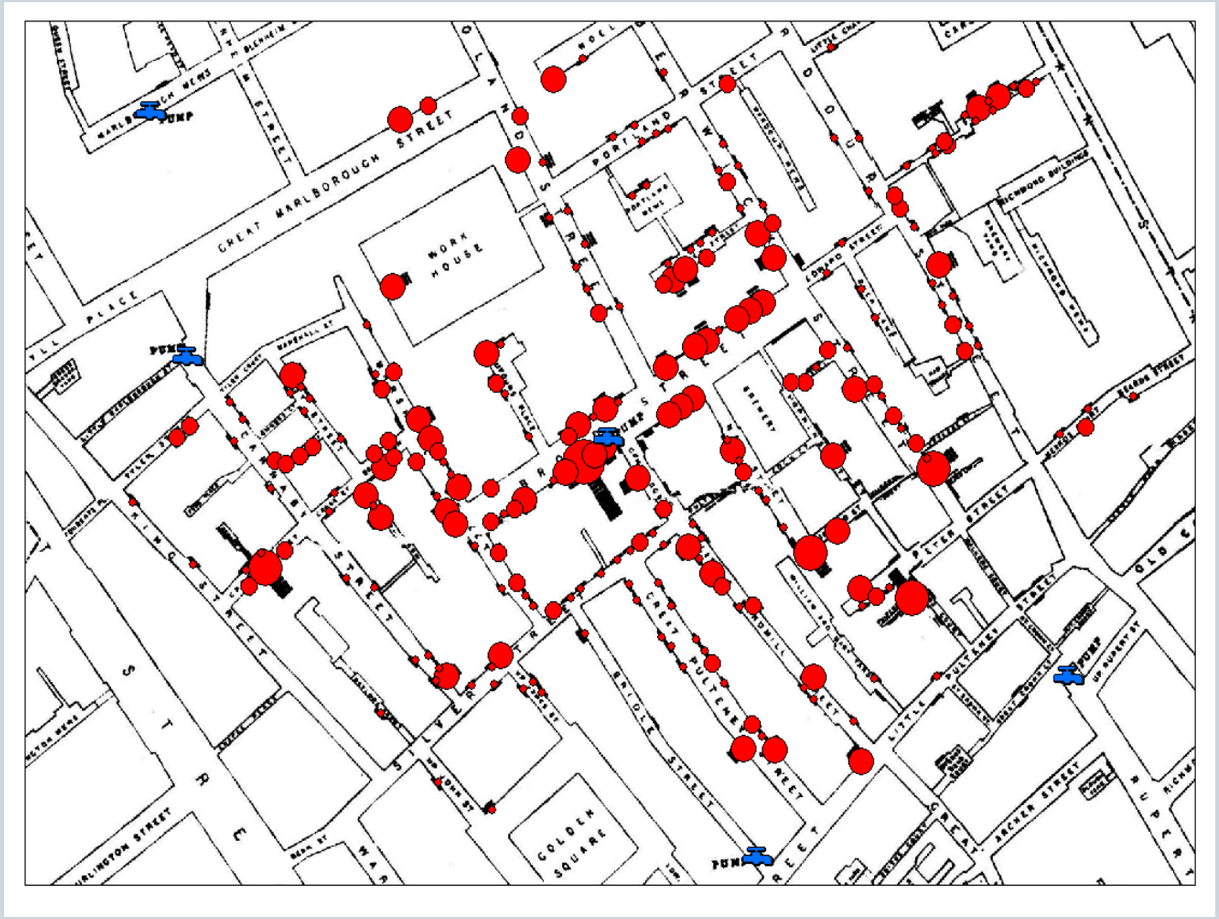
The chart presents a political-economic argument.

The 1854 Soho cholera outbreak

- **Outbreak & Peak**: Began 31 August 1854, peaking in deaths on 1–2 September.
- **Death Toll**: Killed 127 people in the first 3 days, over 500 within 10 days, and 616 Soho residents in total.
- **Public & Medical Response**: Roughly 75% of residents fled; overwhelmed hospitals, used ineffective remedies (opium, brandy, and mercury) while officials poured lime into drains to combat “bad air” (miasma).



John Snow’s argument via data



Each red dot (node) represents accumulated cholera deaths at a recorded address. Larger nodes show more deaths

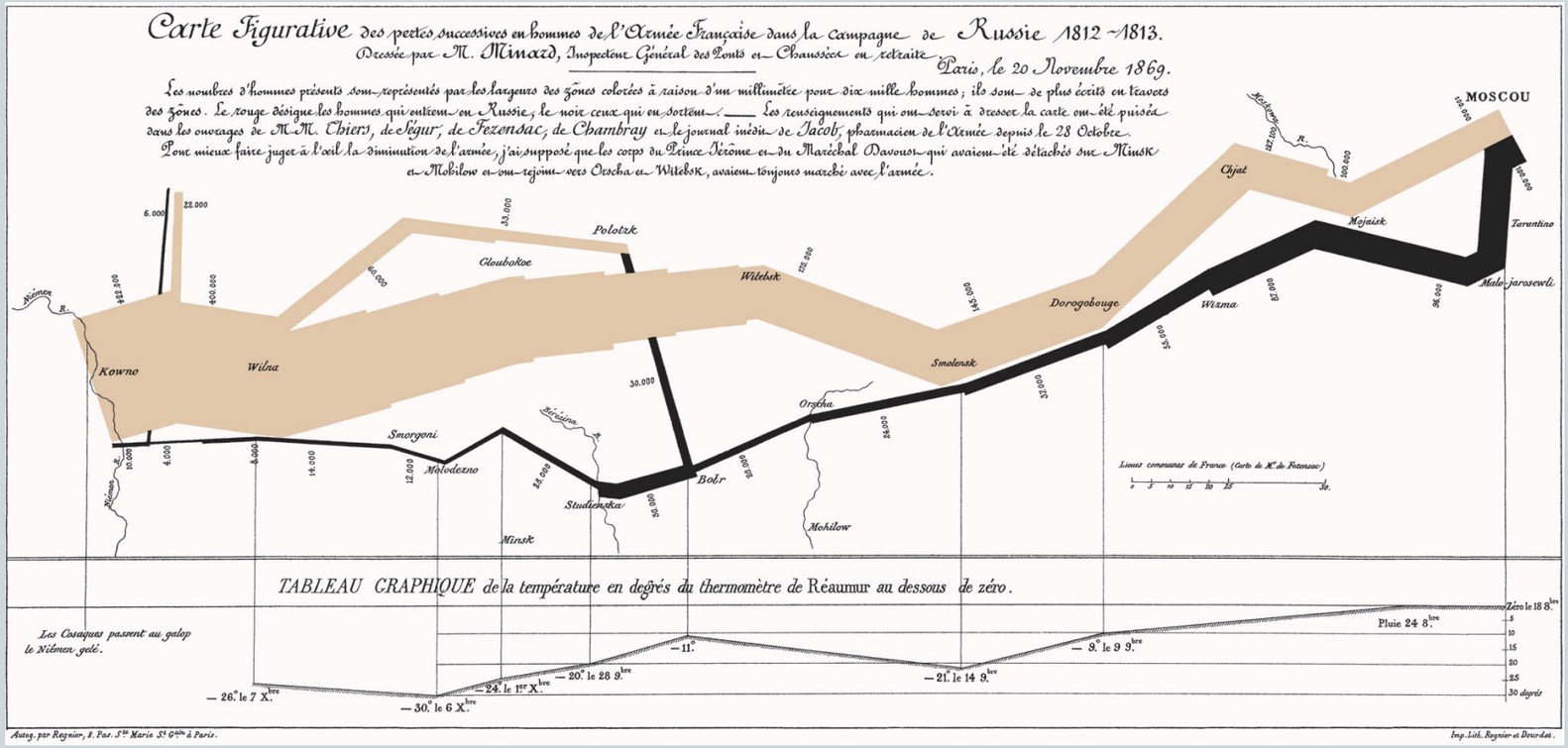
Each blue icon is a pump location.

Deaths cluster around the Broad Street pump, on today’s Broadwick Street.

Snow combined mapped deaths with inquiries about water use. Authorities removed the handle on 8 September.

The outbreak stopped soon after (although it was already declining).

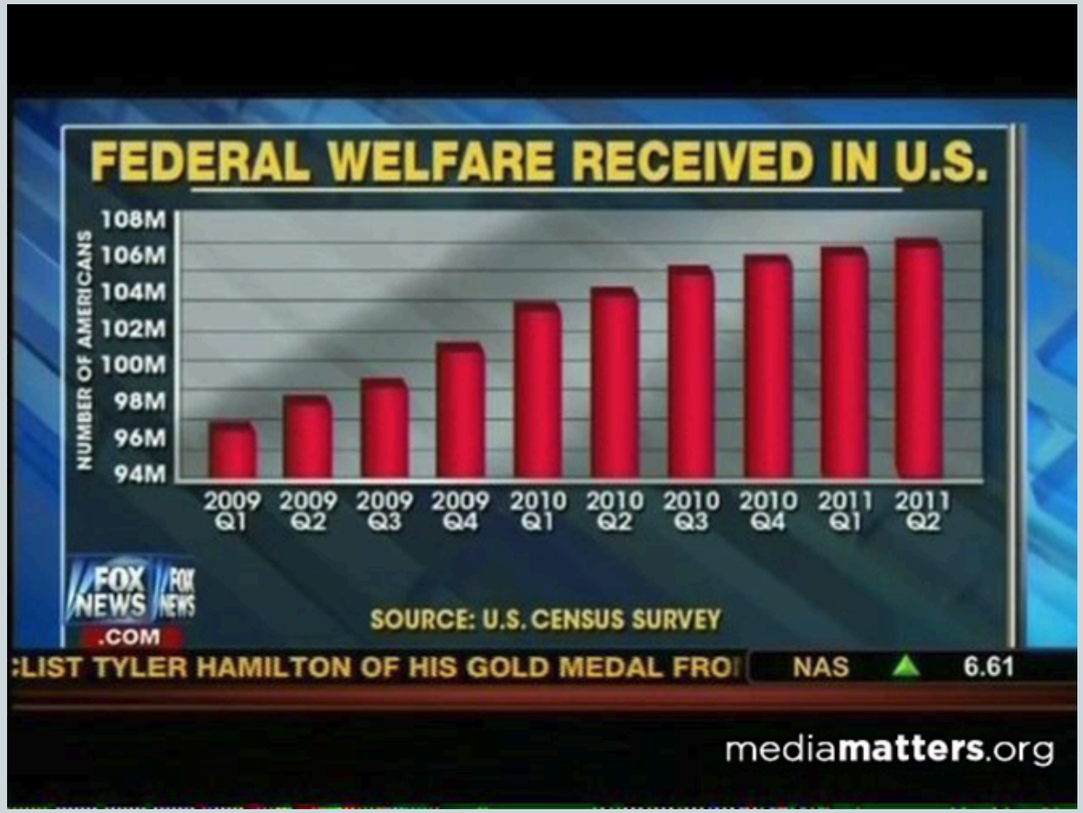
Minard: several variables in one image



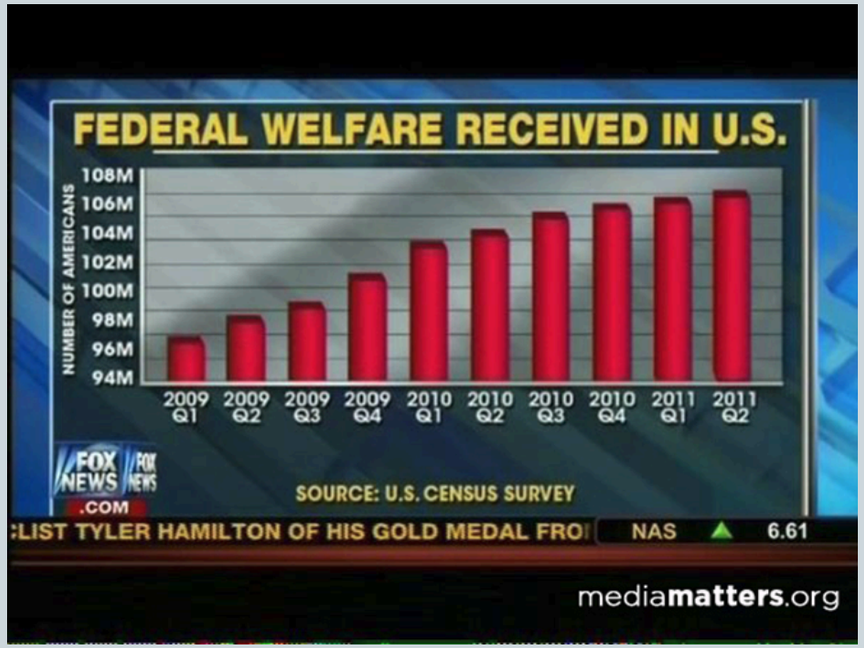
Width: **troop numbers**. Position: **location**. Colour: **outward/return journey**. Dates and temperature accompany the retreat.

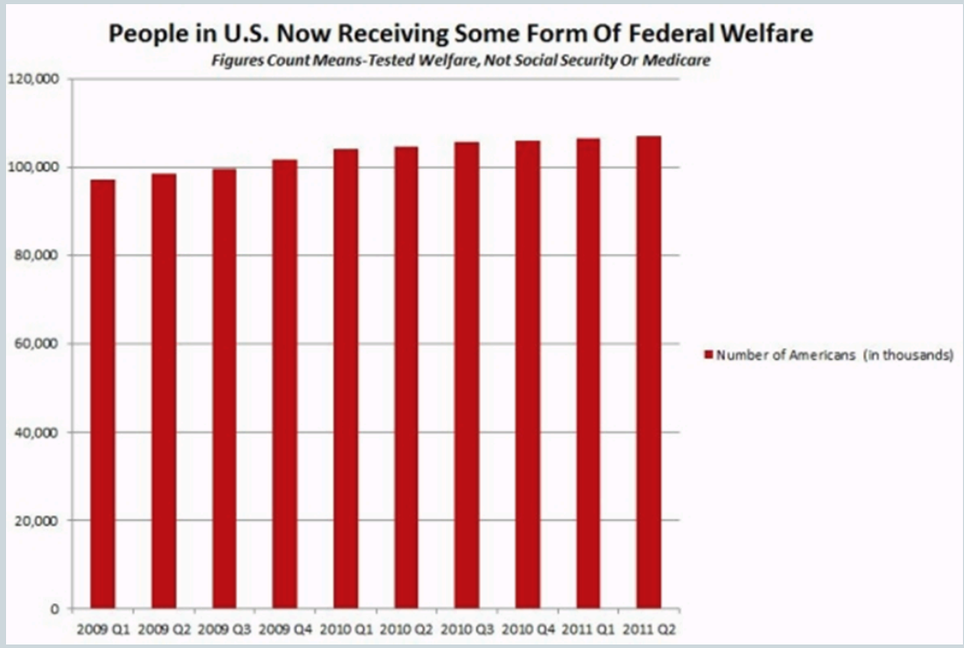
Several variables tell one story, but the reader must learn the encodings. (The temperature scale is **Réaumur**, not Celsius.)

A problematic plot I

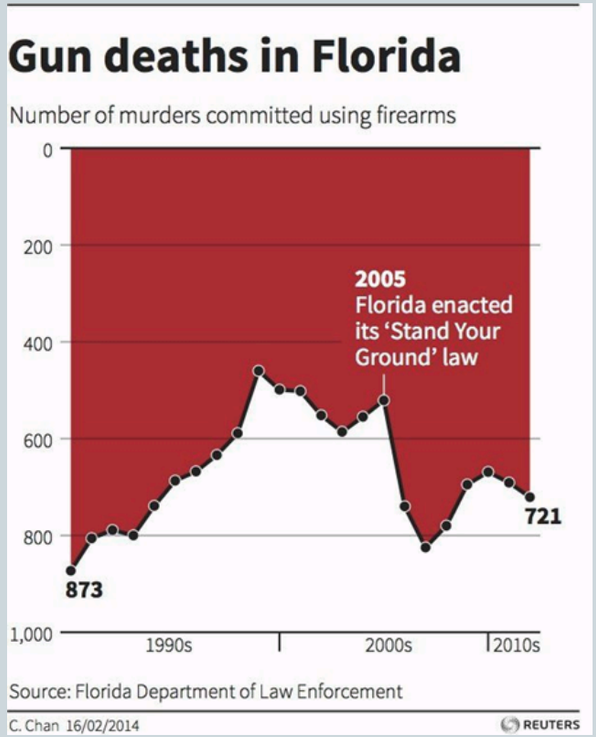


A problematic plot I

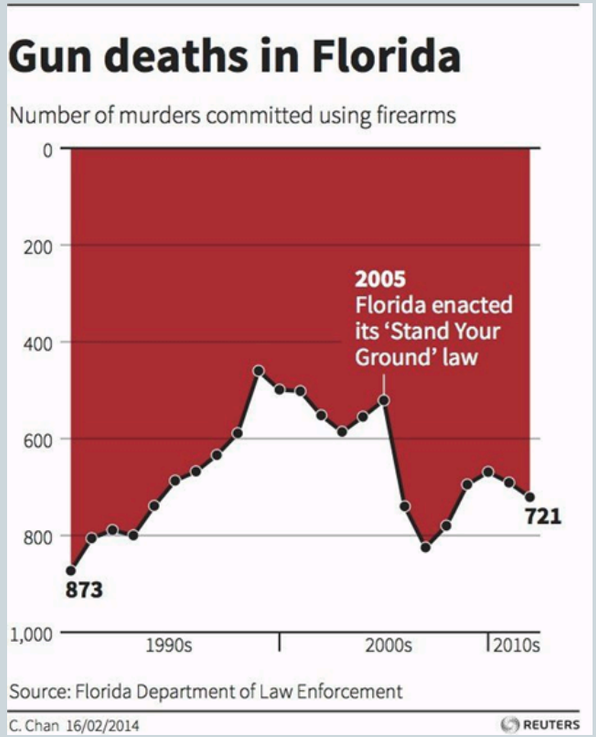


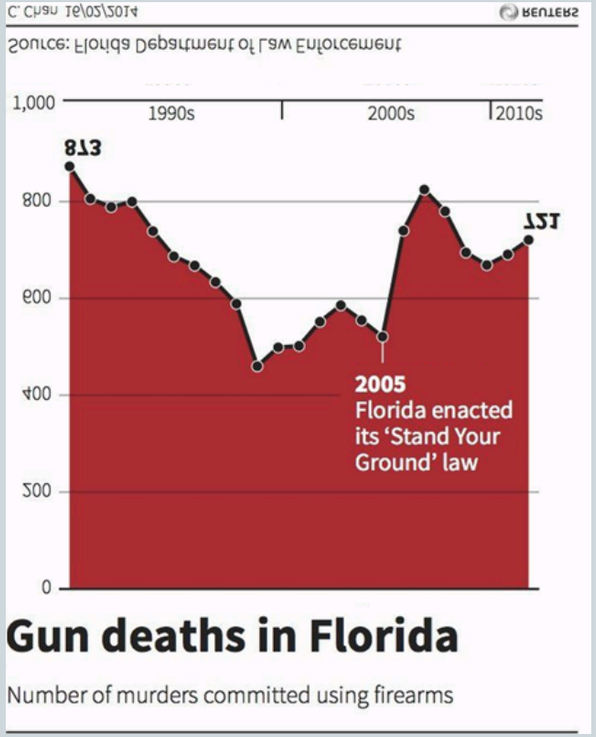


A problematic plot II

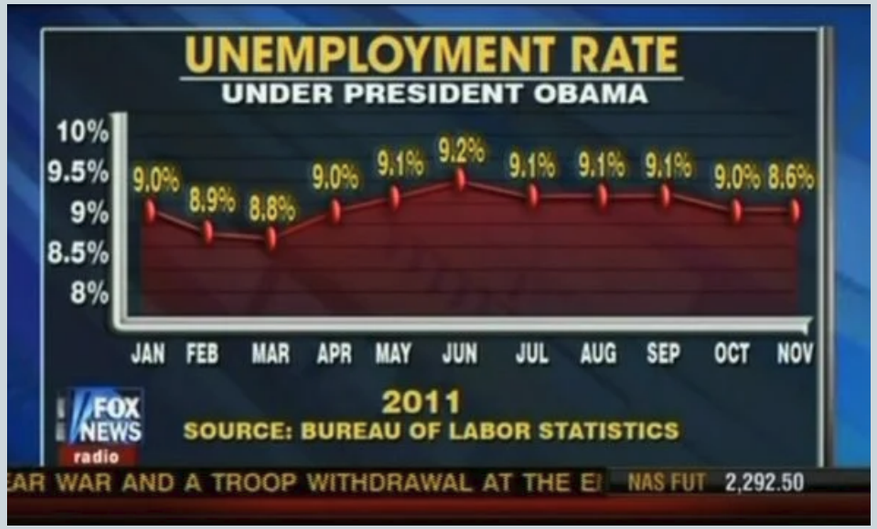


A problematic plot II

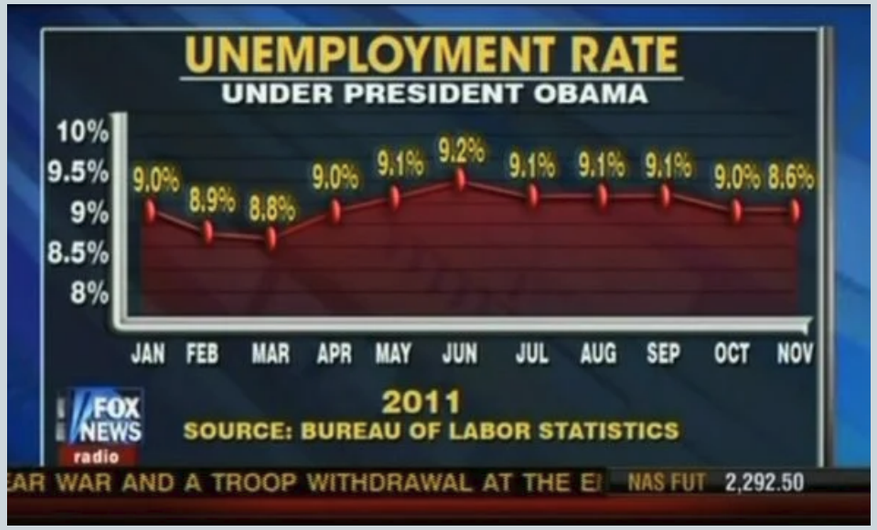


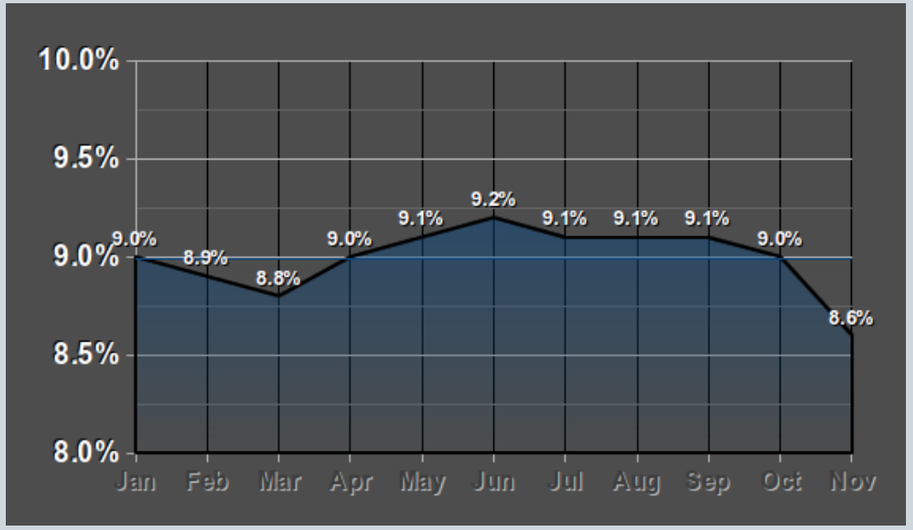


A problematic plot III



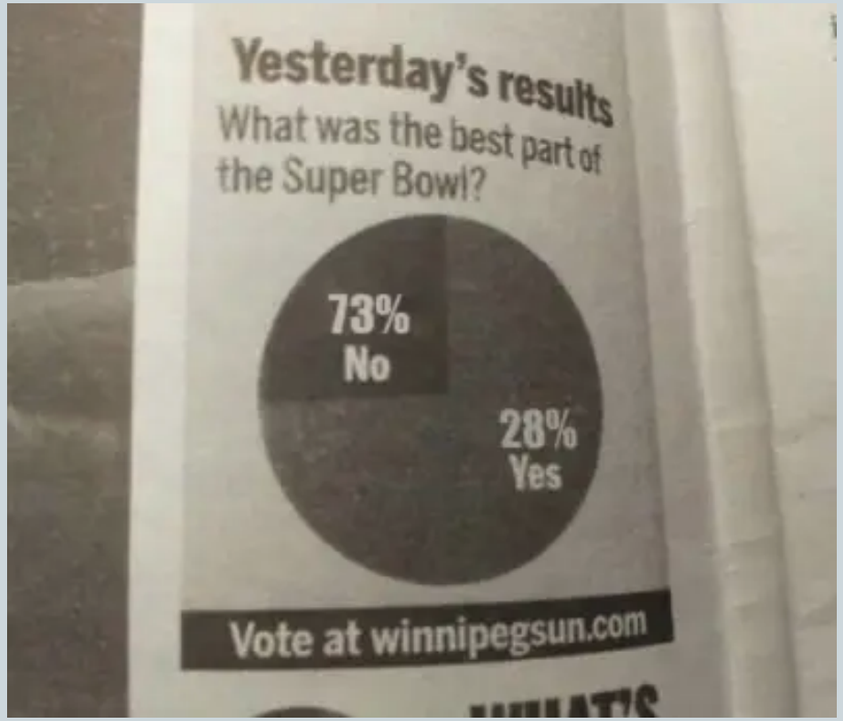
A problematic plot III





On an upward-increasing axis, the final **8.6%** point should be below earlier **8.8%** and **8.9%** points. Check labels against positions, not just apparent line direction.

A very problematic plot



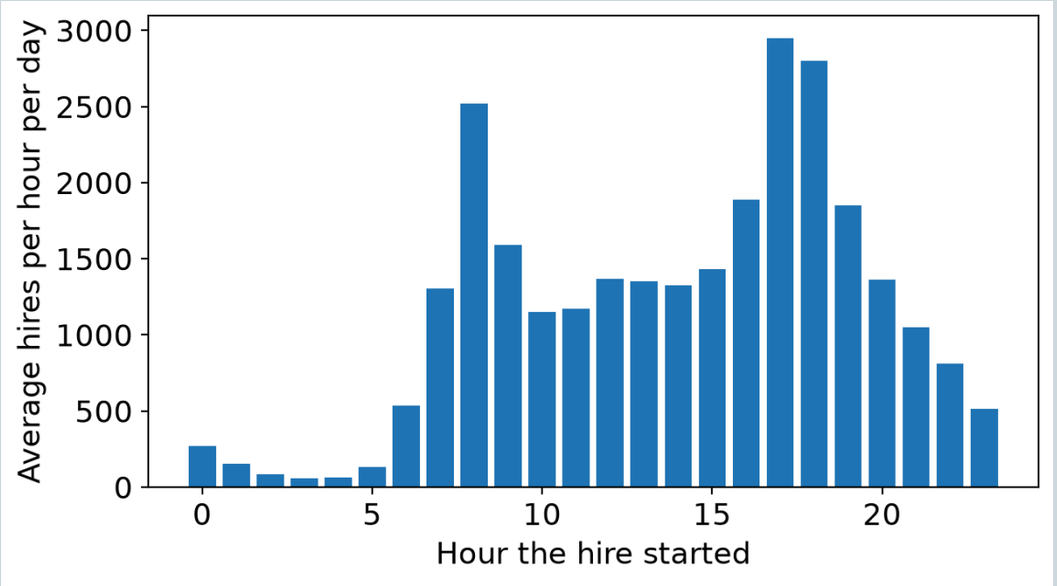
Part II: Line plots

1. **Draw one with `ax.plot()`:** the same figure and axes as bar plots.
2. **Join the dots only when the order is real:** days and hours, not categories.
3. **Put two lines on one axis:** and say what makes them comparable.
4. **Read the shape:** where the line rises, falls, and repeats.

Hires per hour, as bars

Live notebook **01** · Hires per hour, as bars

```
hourly = trips.groupby("hour").size() / 16
fig, ax = plt.subplots(figsize=(7, 4))
ax.bar(hourly.index, hourly.values)
ax.set_xlabel("Hour the hire started")
ax.set_ylabel("Average hires per hour per day")
```

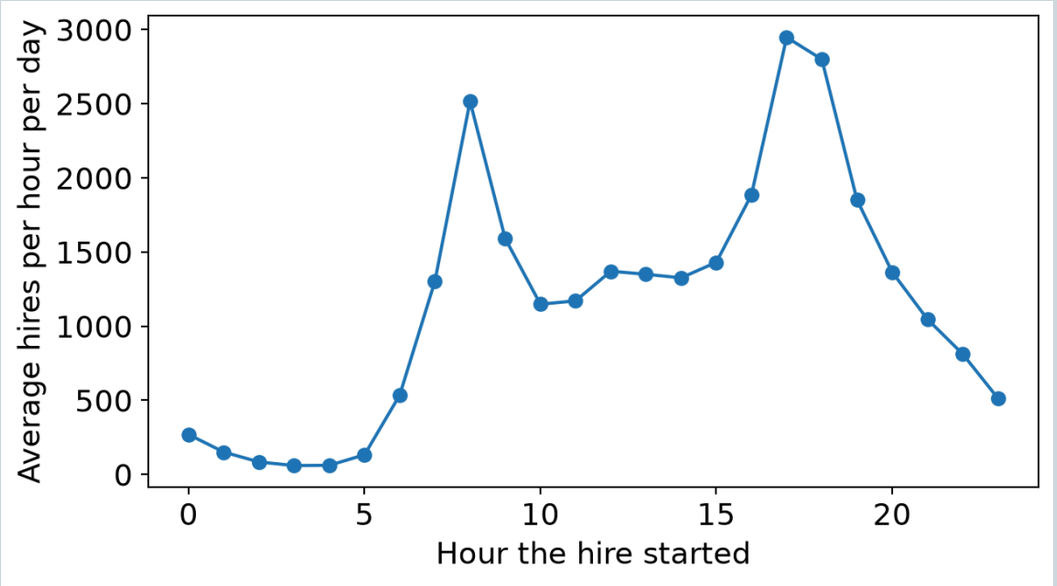


One bar per **start hour**, averaged over **16 dates, 16 to 31 July 2025**. `.size()` counts the hires in each hour; dividing by 16 gives an average per day. Bar lengths are measured from **zero**.

Joined lines vs bars

Live notebook **02** · Joined lines vs bars

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(hourly.index, hourly.values, marker="o")
ax.set_xlabel("Hour the hire started")
ax.set_ylabel("Average hires per hour per day")
```



Same hours, same values. `ax.plot()` draws points and the segments between them. The hour order and the one-hour intervals mean something; the segments help us follow the shape.

The markers show **hourly summaries**, not individual hires or additional observations between hours.

Two lines on one axis

Live notebook **03** · Two lines on one axis

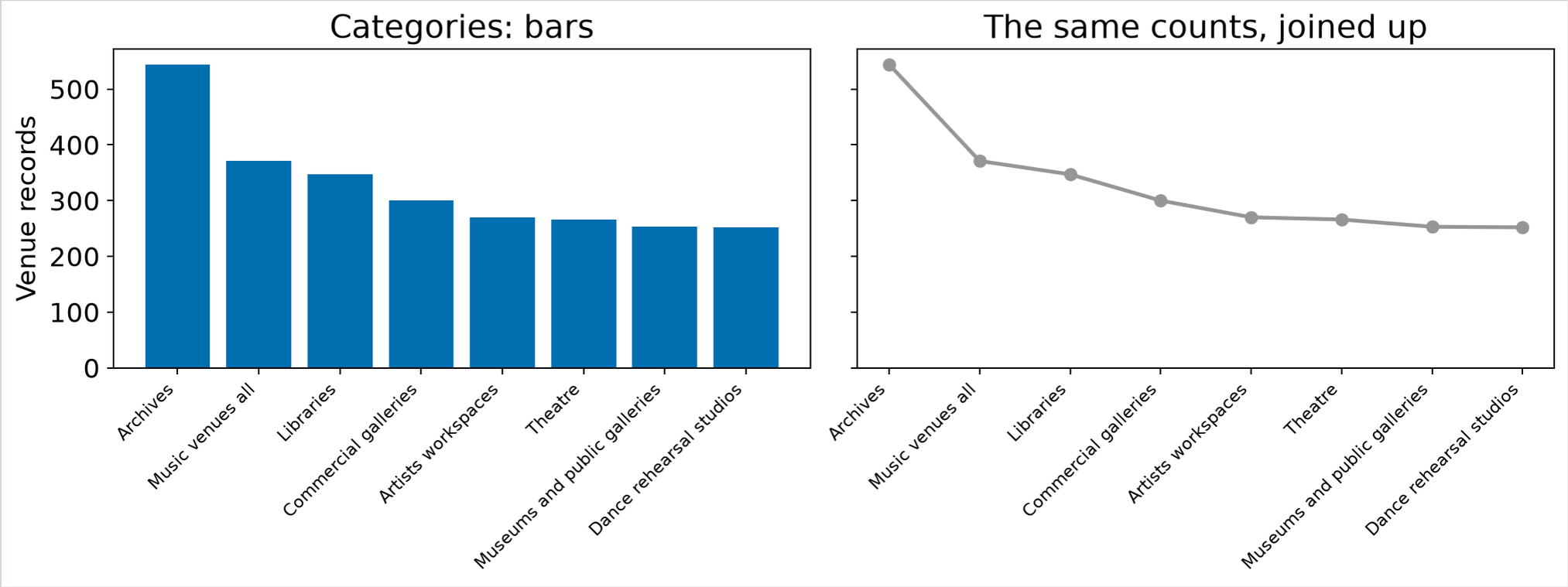
```
# Subset data for weekdays and weekends
weekend_trips = trips[trips["weekday"].isin(["Saturday",
        "Sunday"])]
weekday_trips = trips[trips["weekday"].isin(["Monday",
        "Tuesday", "Wednesday", "Thursday", "Friday"])]
# Normalise the data due to unequal number of days (12
        weekdays vs 4 weekend days)
weekday_hourly = weekday_trips.groupby("hour").size() / 12
weekend_hourly = weekend_trips.groupby("hour").size() / 4
# Plot the results
fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(weekday_hourly.index, weekday_hourly.values,
        label="Weekday")
ax.plot(weekend_hourly.index, weekend_hourly.values,
        label="Weekend")
ax.set_xlabel("Hour the hire started")
ax.set_ylabel("Average hires per hour per day")
ax.legend()
```



Same start hours and hire records, now separated by day type. Weekdays have sharp peaks at **08:00 and 17:00**; weekends have a broader afternoon peak, highest at **16:00**. This is consistent with commuting, but the records do not give journey purposes.

Divide by **12 weekdays** and **four weekend days** to compare an average day of each type. The previous line averaged all 16 dates; these are separate group averages, not parts to add together.

When not to join the dots



Same eight counts. The line suggests a **decline** from Archives to Dance rehearsal studios, but there is nothing between them to decline through. The slope is an artefact of sorting the categories by size.

Lines for ordered axes: time, age, temperature. Bars for categories.

Part III: Scatter plots

1. **Two variables at once:** one point per observation, two measurements each.
2. **Read the pattern first:** direction, strength and form, before any number.
3. **Know the anatomy:** what the axes, the points and the spread each stand for.
4. **See why the picture is not optional:** four datasets with identical statistics.

Why relationships matter

In culture, phenomena rarely exist in isolation:

- Film production budget and later box-office revenue
- Posting time and subsequent social media engagement
- Text length and reading difficulty
- Historical period and genre popularity

Bivariate analysis

Bivariate analysis examines the relationship between two numeric/ordinal variables, asking how they vary together.

Moving beyond describing single variables to understanding how things **relate**.

Key questions

- Do the variables change together?
- When one increases, does the other increase or decrease?
- How strong is the relationship?
- Is there a pattern we can describe?

How we assess

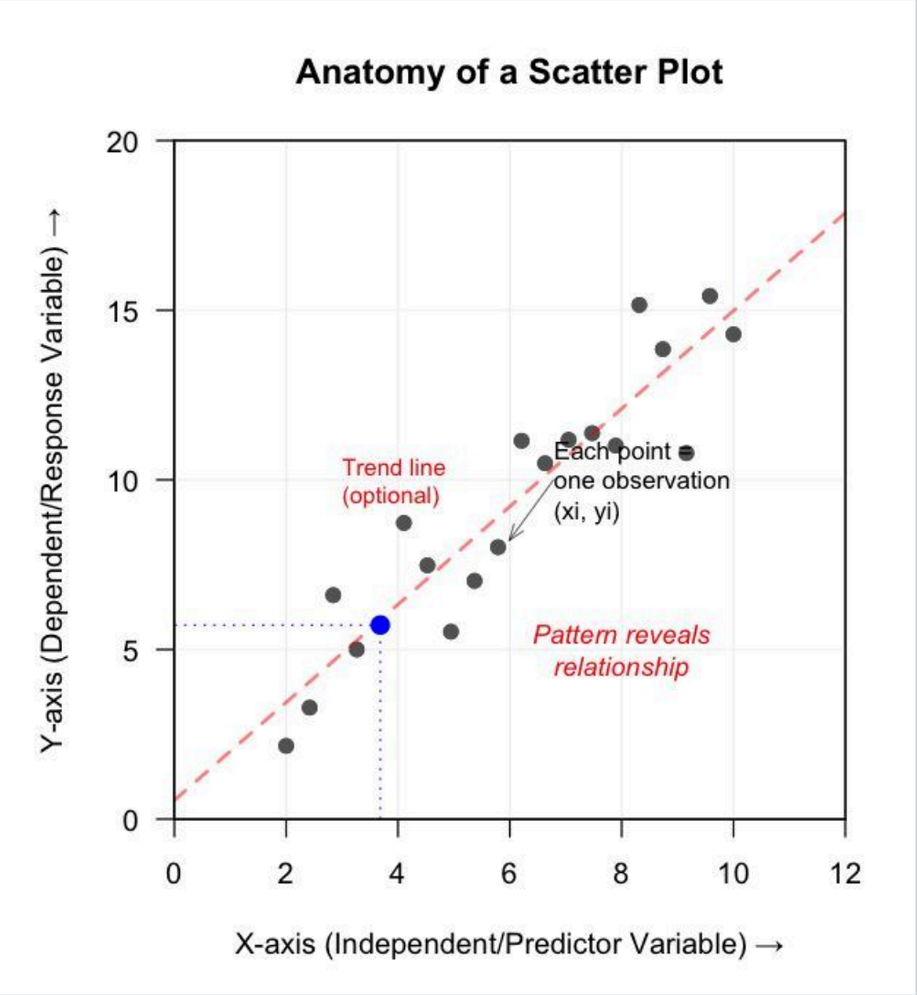
1. **Scatter plots**: each point is one observation; position shows both values; the pattern reveals the relationship, or its absence
2. **Measures of correlation**: Pearson, Spearman, regression models

Anatomy of a scatter plot

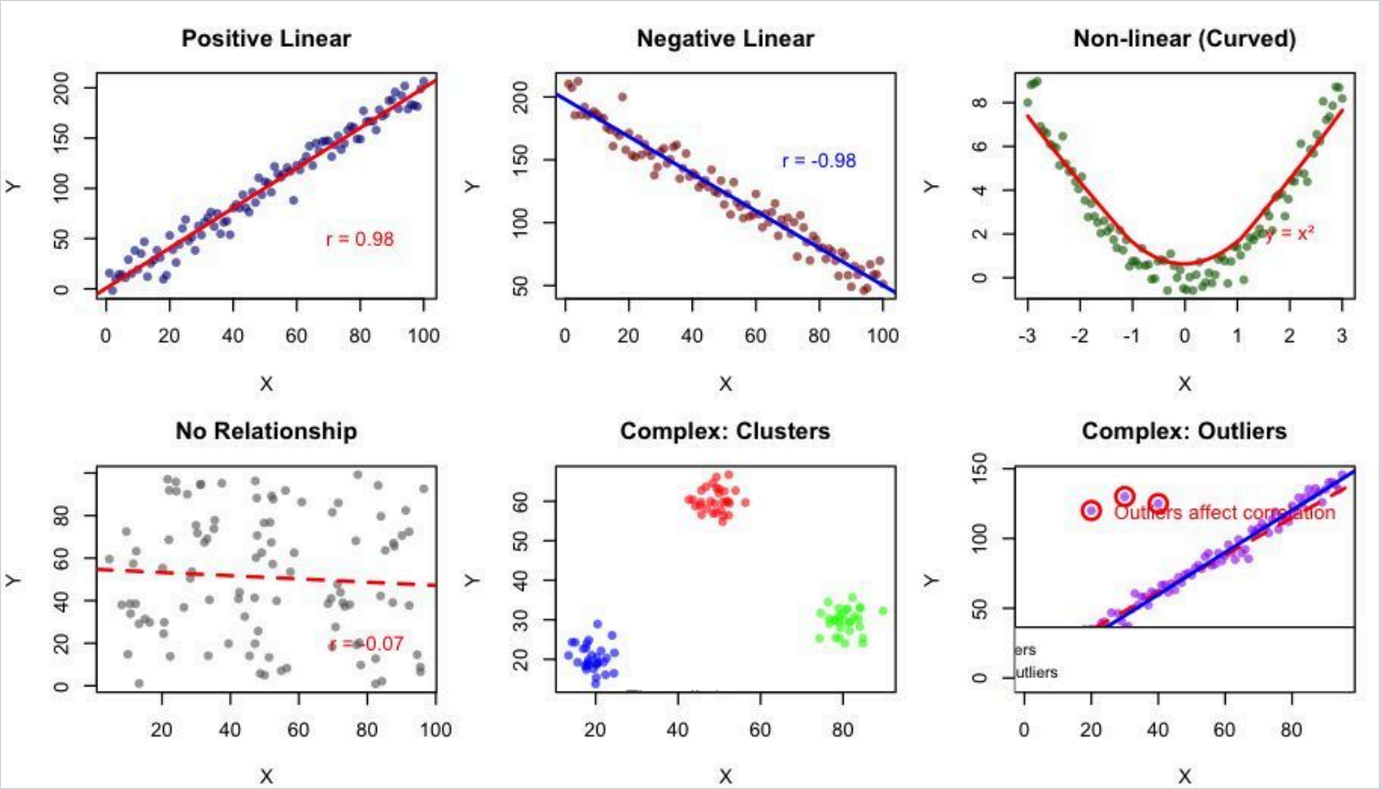
- **X-axis:** independent / predictor variable
- **Y-axis:** dependent / response variable
- **Each point:** one observation
- **Pattern:** reveals the relationship

When to use

- To inspect a potential relationship between two numeric variables
- To verify statistical analyses: correlation, linear regression



Types of relationships



r is the correlation coefficient: one number summarising the direction and strength of a straight-line pattern, from −1 through 0 to +1. Part IV calculates it.

Part IV: Correlations

1. **Put a number on the pattern:** strength and direction, between −1 and +1.
2. **Choose a coefficient:** Pearson on values, Spearman on ranks.
3. **Calculate both in pandas:** with `.corr()`, and state the n they describe.
4. **Interpret with care:** two variables moving together is not one causing the other.

What is correlation?

Correlation measures two things:

1. **Strength** of association between two variables
2. **Direction** of the relationship

Coefficients range from **−1 to +1**:

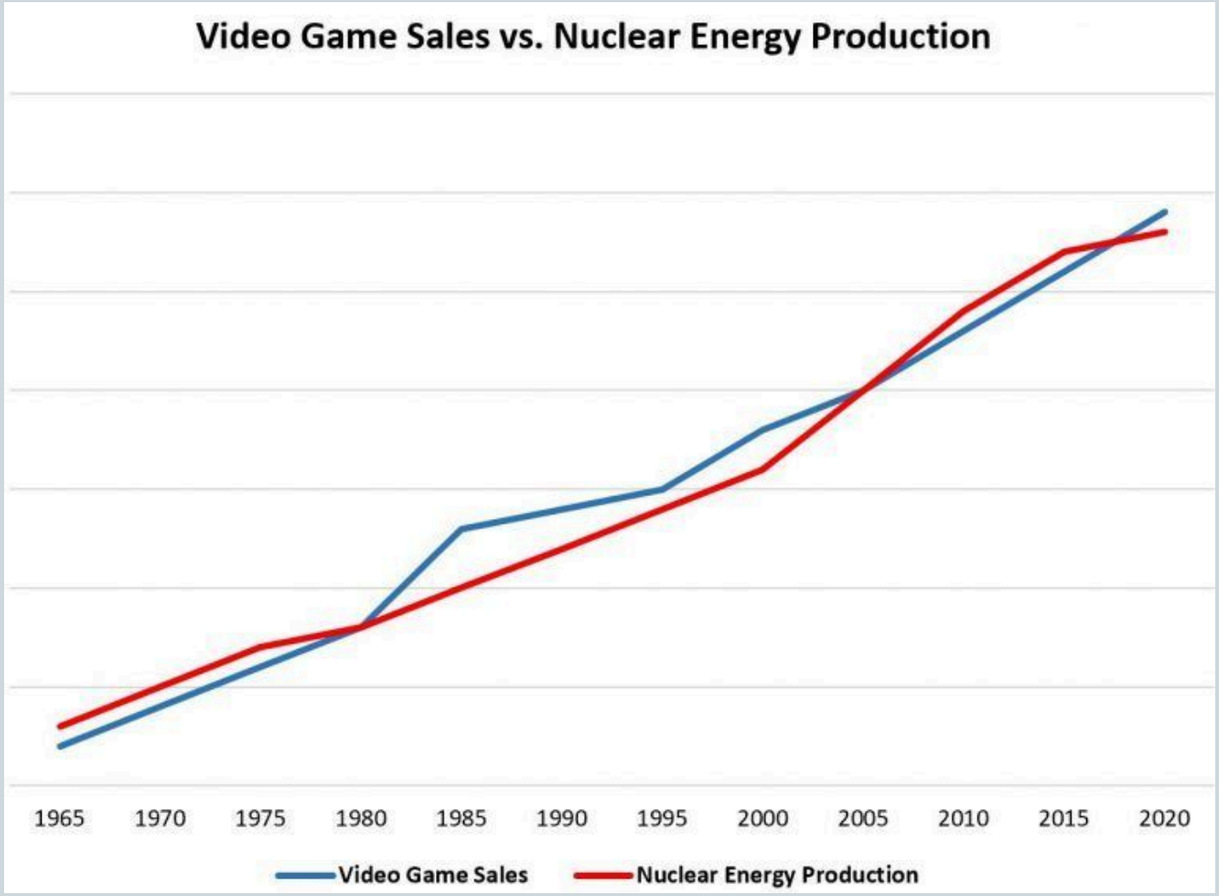
- ±1 = perfect straight-line association
- near 0 = little or no **linear** association (could be some other relationship)
- > 0 = positive relationship and < 0 = negative relationship

Correlation ≠ Causation

A coefficient says two variables **move together**. It does not say which one moves the other, or whether something else moves both.

For example, the number of venues in a borough goes up along with the overal public transport accessibility. That relationships, however, cannot tell us whether *transport* encouraged venues to open, or the *abundance of venues* resulted in transport opening (or some other, unmeasured variable impacting both).

Bad correlations



Tyler Vigen collects these:
https://tylervigen.com/spurious-correlations

- US cheese consumption and deaths by bedsheet entanglement, **r = 0.95**
- Films Nicolas Cage appeared in and drownings in swimming pools, **r = 0.67**

Enough series, enough years, and something will match. A high coefficient on its own is not evidence of anything.

Interpreting strength

r		Interpretation
0.8 to 1	−0.8 to −1	Very strong
0.6 to 0.8	−0.6 to −0.8	Strong
0.4 to 0.6	−0.4 to −0.6	Moderate
0.2 to 0.4	−0.2 to −0.4	Weak
0 to 0.2	0 to −0.2	Very weak / none

Note: the importance of an r value differs depending on the data. In some fields 0.3 is a major finding; in others 0.8 is unremarkable.

Pearson correlation

Measures the strength of a **straight-line** relationship.

- Uses your actual data values
- Best for “as X increases, Y increases in a straight line”

Strength: very precise when the relationship is linear

Weaknesses: one extreme value can throw it off badly; misses curves

When to use

- Both variables continuous
- The scatter plot looks roughly like a straight line
- No wild outliers

If you could draw a straight line through it, use Pearson.

Spearman correlation

Measures whether two variables **consistently move together**.

- Converts data to **rankings** first
- Captures any consistent pattern, including even curves

Strengths: handles curves; outliers do not upset it; works with ordered categories such as Likert scales

Weakness: slightly less powerful than Pearson when the relationship really is linear

When to use

- Ordinal data (rankings, survey scales)
- Curved but consistent relationships
- Outliers present
- Unsure about the distribution

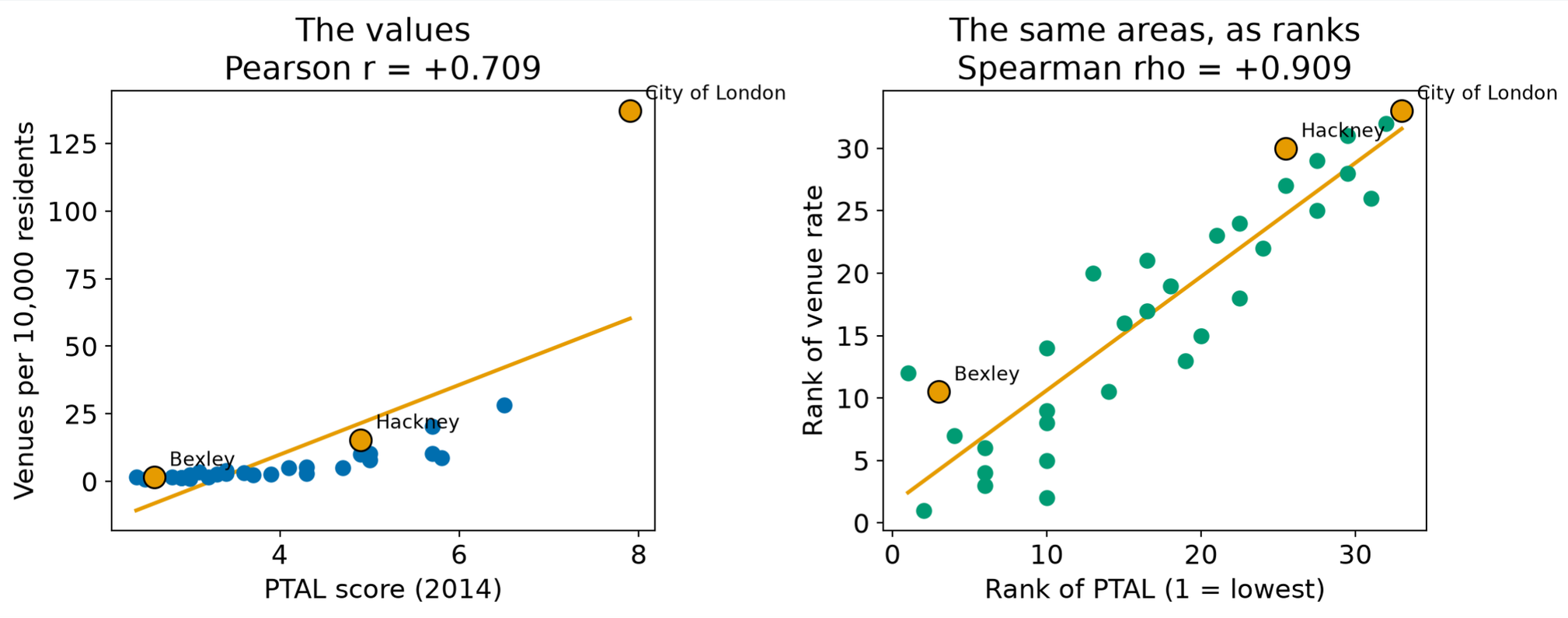
Consistent movement, but curvy or with outliers, use Spearman.

What do we mean by rankings?

Original X	Ranked X	Original Y	Ranked Y
1	1	2	1
3	2	5	2
7	3	10	3
10	4	18	4
12	5	30	5
14	6	35	6
15	7	45	7

Spearman correlates the **ranks**, not the values. So it does not care how far apart the numbers are, only about their order.

The same areas, as values and as ranks



Left: the values. **Right**: the same 33 areas after both columns are replaced by their ranks.

Watch the City of London. On the left it is 137.04 against a median of 2.81, miles off the line. On the right it can only be **rank 33 of 33**, one place above its neighbour, and it sits near the line like everything else.

Correlation in Python

Live notebook **04** · Correlation in Python

```
1  pair = boroughs[["ptal_score_2014", "venues_per_10k_residents"]].dropna()
2  x = pair["ptal_score_2014"]
3  y = pair["venues_per_10k_residents"]
4  print("n =", len(pair))
5  print("Pearson:", round(x.corr(y), 3))
6  print("Spearman:", round(x.corr(y, method="spearman"), 3))
```

```
n = 33
Pearson: 0.709
Spearman: 0.909
```

`.corr()` is a **Series method**: call it on one column and pass the other as its argument. It returns a single number, Pearson by default, and `method="spearman"` asks for the other. The order does not matter: `x.corr(y)` and `y.corr(x)` are the same value.

One row per borough area. `.dropna()` first, so both coefficients describe the same **n = 33**.

When Pearson and Spearman disagree

```
Pearson : 0.709
Spearman: 0.909
```

> Pearson says 0.709. Spearman says 0.909. Same 33 areas, same two columns. Why the gap?

Pearson measures how close the points sit to a straight line. Spearman replaces each value with its **rank** and measures the same thing on those. A few areas far above the rest weaken the first and barely touch the second.

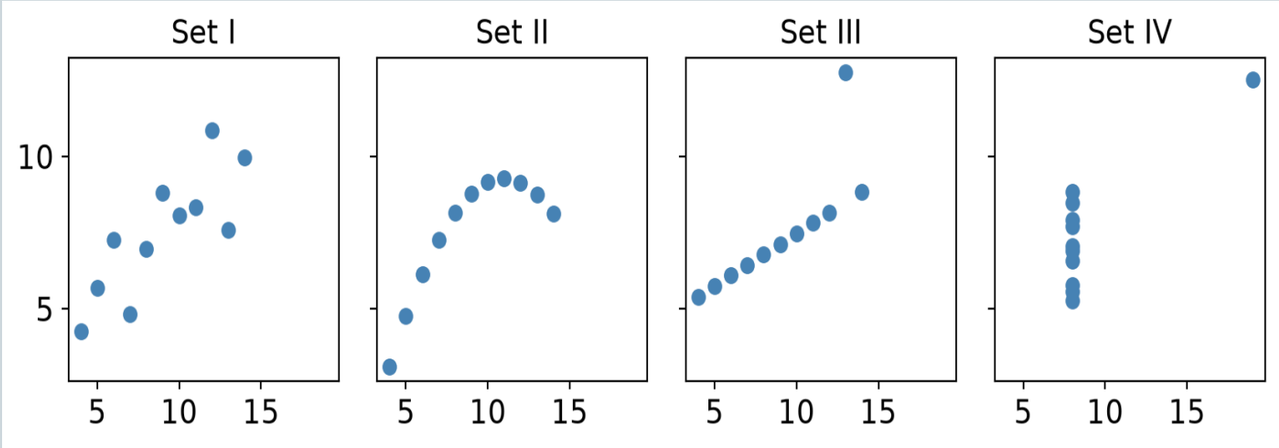
Why visualise, when we can measure?

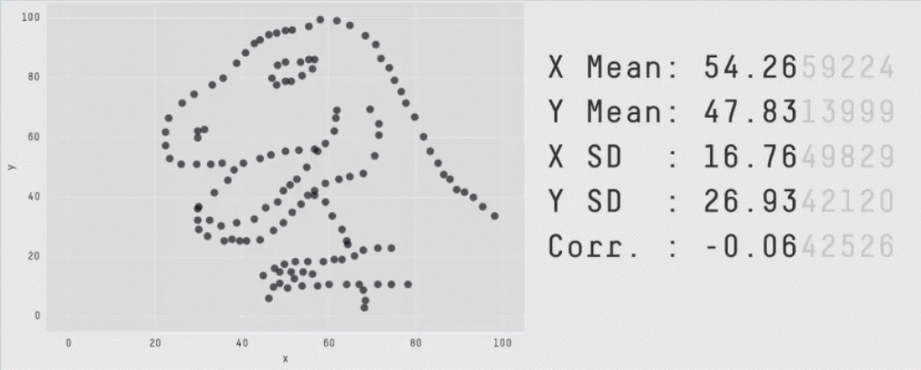
Because two datasets can have identical statistics and completely different shapes.

Anscombe’s quartet

Four datasets, **11 observations each**. Means, **sample SDs** and Pearson’s **r**:

set	mean x	sd x	mean y	sd y	r
I	9.00	3.32	7.50	2.03	0.816
II	9.00	3.32	7.50	2.03	0.816
III	9.00	3.32	7.50	2.03	0.816
IV	9.00	3.32	7.50	2.03	0.817





Datasaurus Dozen: changing shapes, summaries preserved to **two decimal places**.

Almost identical summaries. Different shapes. Plot the data.

Anscombe (1973); animation: Matejka and Fitzmaurice (2017), *Same Stats, Different Graphs*.

Part V: Drawing scatter plots that work

1. **Choose the channels:** position, size, colour, shape and transparency.
2. **Add a third variable:** when it answers a question the first two raised.
3. **Handle too many points:** smaller markers and transparency.
4. **Save the result:** a labelled file fit for a report.

Visual variables (Bertin’s semiology)

A scatter plot has more than two channels. Each one can carry information, and some carry quantity better than others.

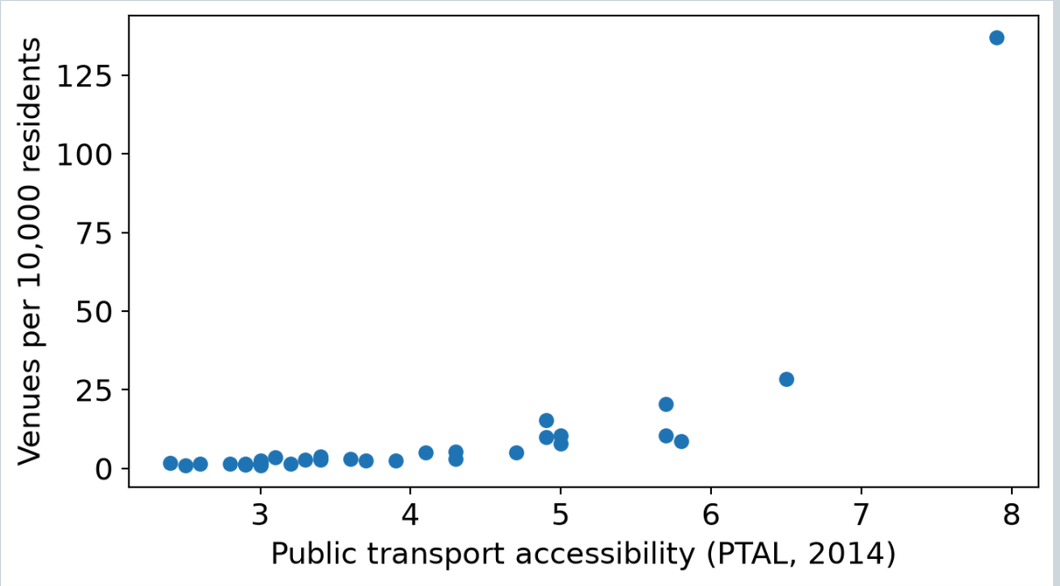
Visual variable	Best for	Example
Position (x, y)	the two variables you are relating	rate against accessibility
Size	a third numeric variable	resident population
Colour (saturation)	a continuous value	intensity, amount
Shape	a few categories	different sources
Transparency	overlapping points	density, where there are too many to see

Monday used the same idea for bars: **length** carried the number and **hue** separated groups. Hue still separates groups here, and still does not say which is larger.

A basic scatter plot

Live notebook **05** · A basic scatter plot

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(boroughs["ptal_score_2014"],
           boroughs["venues_per_10k_residents"])
ax.set_xlabel("Public transport accessibility (PTAL,
           2014)")
ax.set_ylabel("Venues per 10,000 residents")
```



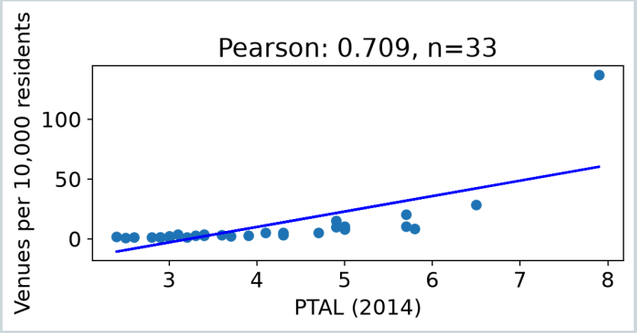
One point per borough area, 33 in all. Better-connected areas tend to have higher recorded rates, and one point sits a long way above the rest. Next, add Pearson and Spearman to this figure.

Add Pearson and Spearman

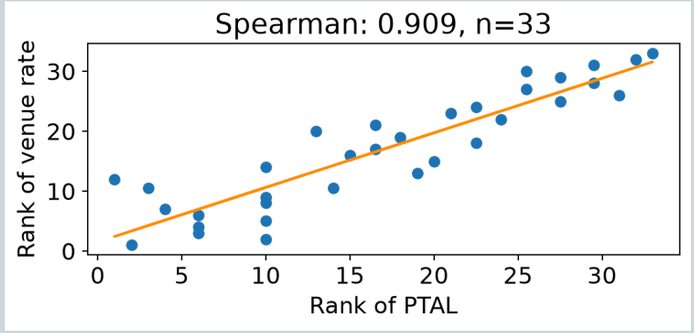
Live notebook **06** · Add Pearson and Spearman

```
pair = boroughs[["ptal_score_2014", "venues_per_10k_residents"]].dropna()
x = pair["ptal_score_2014"]
y = pair["venues_per_10k_residents"]
```

```
pearson = round(x.corr(y), 3)
fig, ax = plt.subplots(figsize=(6, 3))
ax.scatter(x, y)
slope, intercept = np.polyfit(x, y, 1)
ax.plot(x, slope * x + intercept, color="blue")
ax.set_xlabel("PTAL (2014)")
ax.set_ylabel("Venues per 10,000 residents")
ax.set_title(f"Pearson: {pearson}, n={len(x)}")
scatter_fig = fig
```




```
rx = x.rank()
ry = y.rank()
spearman = round(x.corr(y, method="spearman"), 3)
fig, ax = plt.subplots(figsize=(6, 3))
ax.scatter(rx, ry)
slope, intercept = np.polyfit(rx, ry, 1)
ax.plot(rx, slope * rx + intercept, color="darkorange")
ax.set_xlabel("Rank of PTAL")
ax.set_ylabel("Rank of venue rate")
ax.set_title(f"Spearman: {spearman}, n={len(x)}")
```

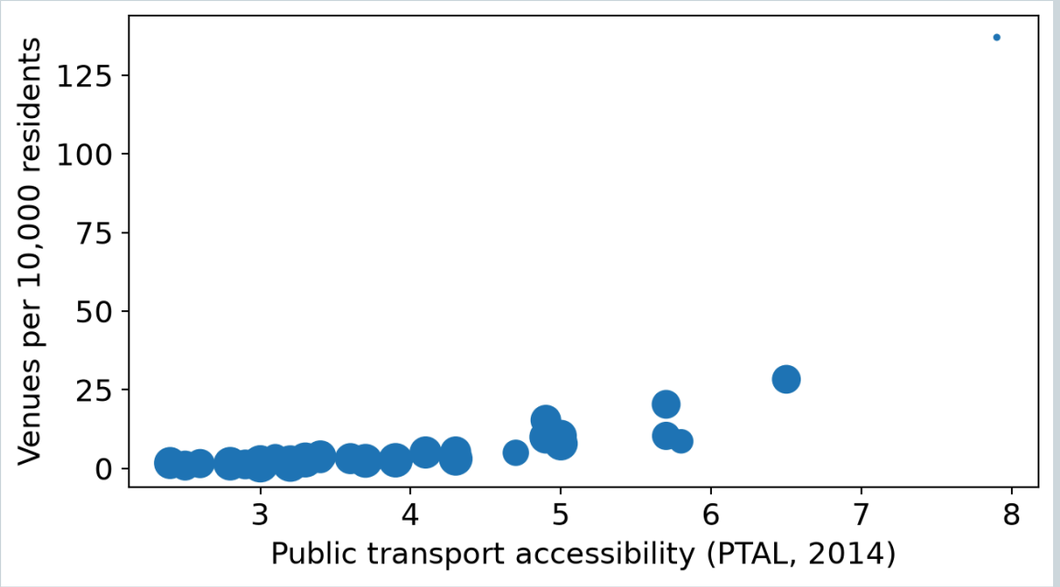


`np.polyfit(..., 1)` finds the slope and intercept. `ax.plot(...)` draws the line. The orange line uses ranked values, because Spearman is calculated on ranks.

A third variable: size

Live notebook **07** · A third variable: size

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(boroughs["ptal_score_2014"],
           boroughs["venues_per_10k_residents"],
             s=boroughs["population_2015"] / 1500)
ax.set_xlabel("Public transport accessibility (PTAL,
           2014)")
ax.set_ylabel("Venues per 10,000 residents")
```



Marker area = resident population / 1500. Populations run from **8,100 to 393,200**, and the point at the top is now the smallest circle. A circle twice as wide has four times the area.

Too many points to read

Live notebook **08** · Too many points to read

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(within_hour["start_hour"],
           within_hour["duration_min"],
             s=6)
ax.set_xlabel("Hour the hire started")
ax.set_ylabel("Hire duration (minutes)")
```

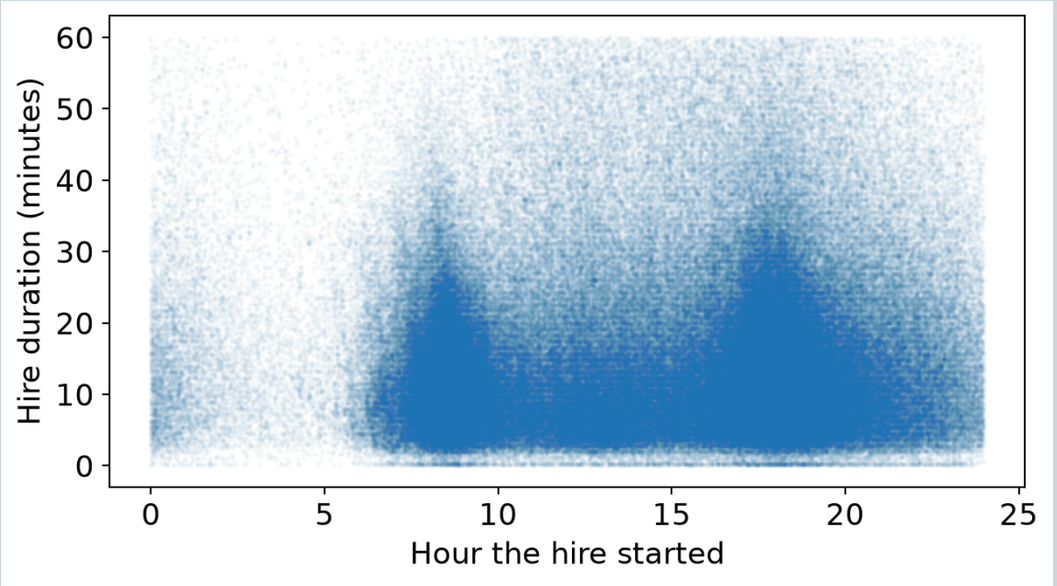


432,667 hires in Monday's 0 to 60 minute window. Every marker is drawn; almost none can be seen. Pearson is **+0.055**.

Making a dense scatter readable

Live notebook **09** · Making a dense scatter readable

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(within_hour["start_hour"],
           within_hour["duration_min"],
             s=2, alpha=0.02)
ax.set_xlabel("Hour the hire started")
ax.set_ylabel("Hire duration (minutes)")
```

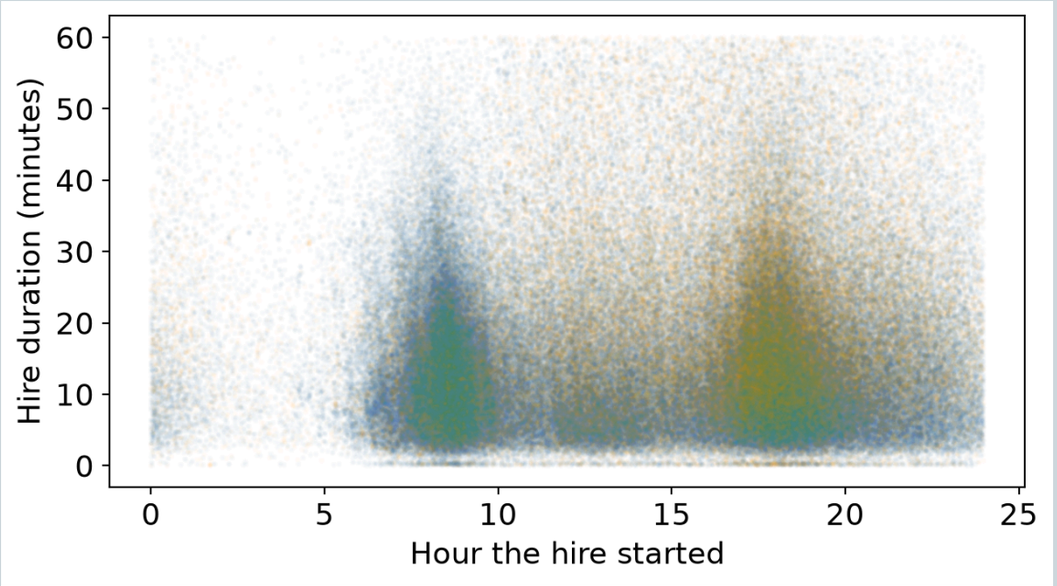


Smaller markers and **2% opacity**. Same hires, same code otherwise. Two peaks appear at roughly **08:00** and **17:30**, and a band of very short hires along the bottom.

A third variable: colour

Live notebook **10** · A third variable: colour

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(weekday["start_hour"], weekday["duration_min"],
           c=weekday["origin_colour"], s=2, alpha=0.02)
ax.set_xlabel("Hour the hire started")
ax.set_ylabel("Hire duration (minutes)")
```

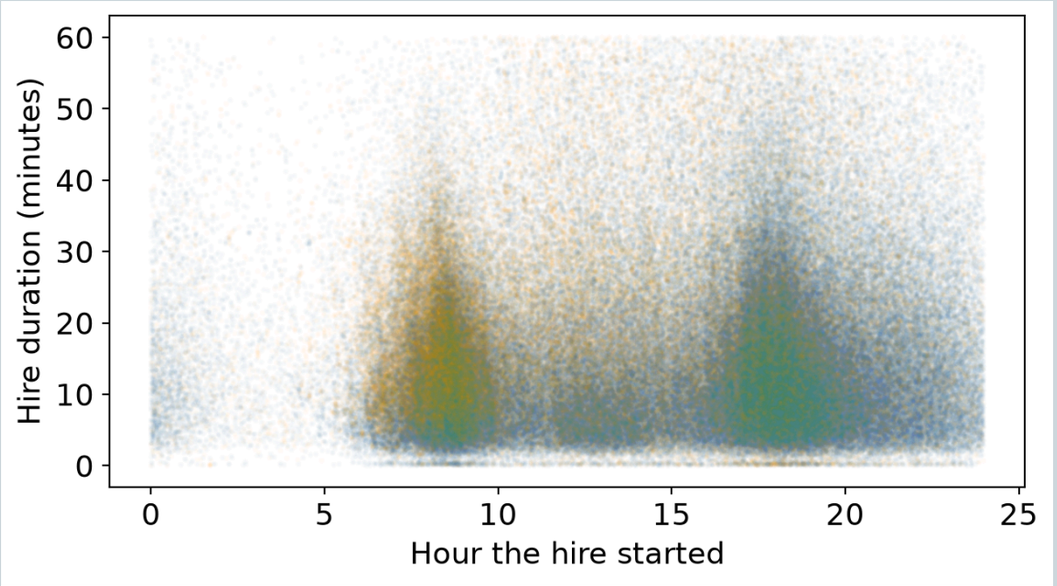


Blue = elsewhere; orange = City or Westminster. Weekday hires only, split by where the hire **started**: **112,133** in the City or Westminster, **238,852** elsewhere. The morning peak is mostly blue and the evening peak mostly orange.

The same hires, coloured by destination

Live notebook **11** · The same hires, coloured by destination

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(weekday["start_hour"], weekday["duration_min"],
           c=weekday["destination_colour"], s=2,
          alpha=0.02)
ax.set_xlabel("Hour the hire started")
ax.set_ylabel("Hire duration (minutes)")
```



Blue = elsewhere; orange = City or Westminster. The previous figure coloured by where the hire **started**. This one colours by where it **ended**, and the peaks swap.

Weekday hires	07:00	17:00
Starting in the centre	20.4%	42.3%
Ending in the centre	47.0%	26.6%

Save the figure from the notebook

Live notebook **12** · Save the figure from the notebook

```
1  scatter_fig.savefig("borough_scatter.png", dpi=200)
2  scatter_fig.savefig("borough_scatter.svg")
```

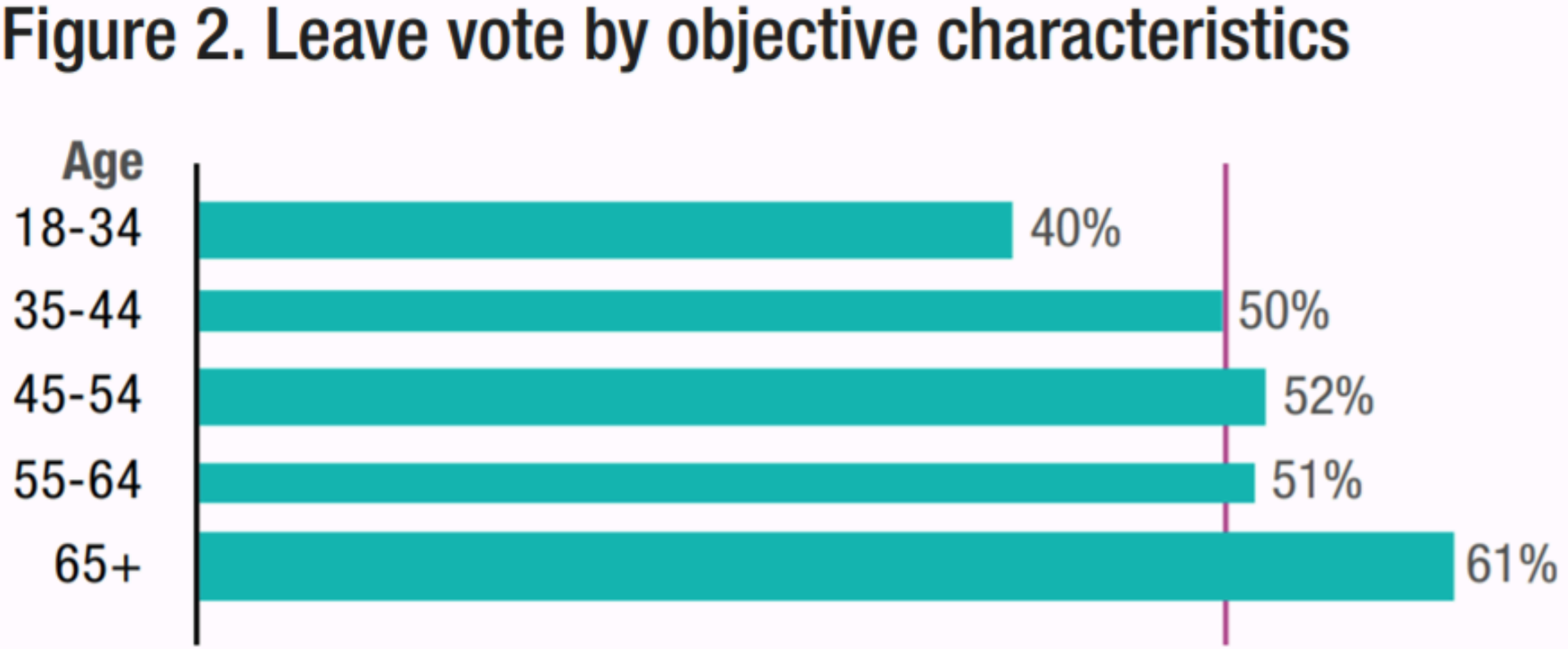
PNG: raster for slides/web. 6 × 3 inches at 200 dpi gives 1,200 × 600 pixels.

SVG/PDF: scalable vector linework and text. Open exports to check labels. `savefig()` writes the file; no separate device-closing step is required.

Save the **figure object**, not a screenshot of the notebook, and keep the code that made it next to the file.

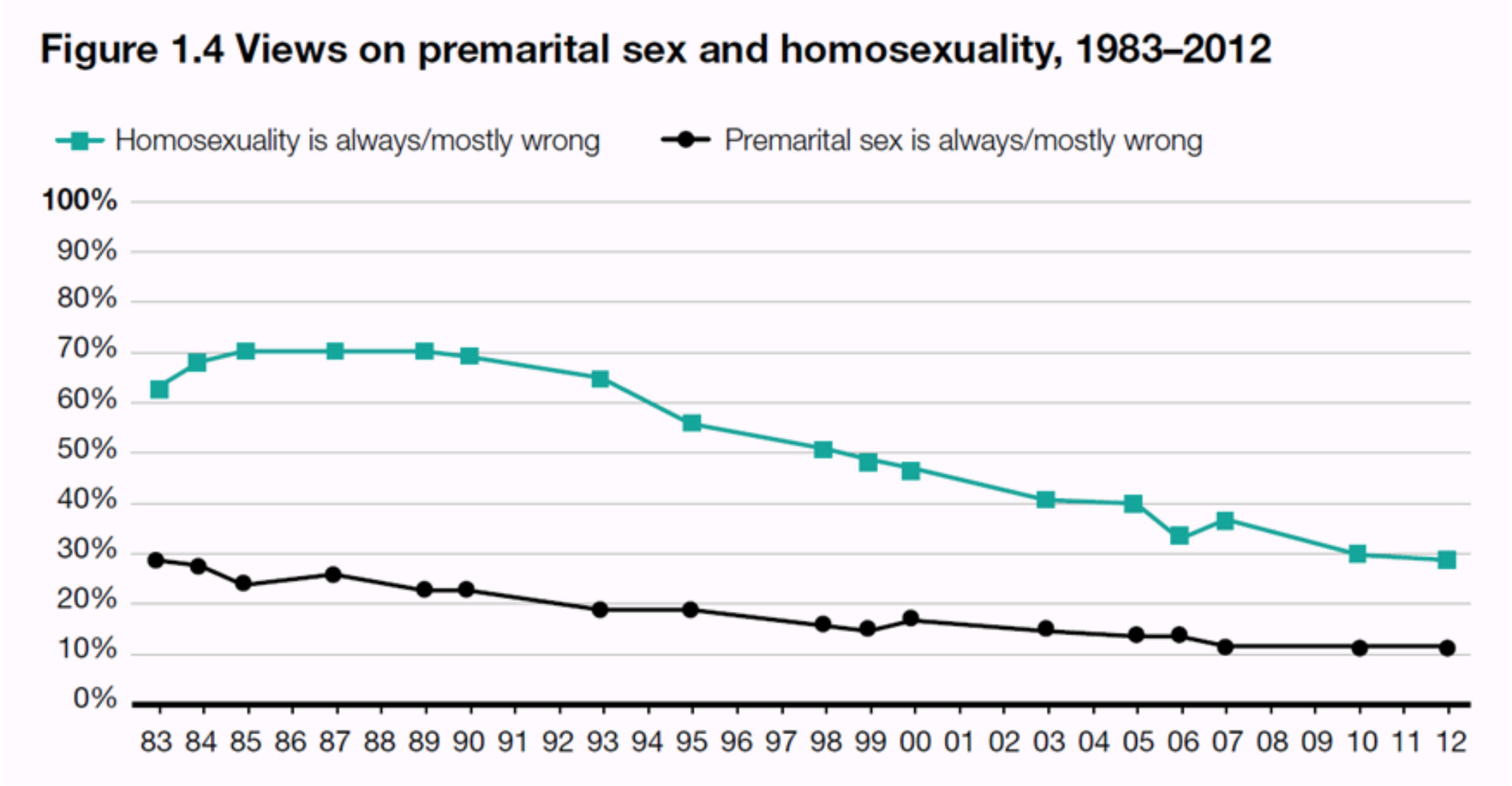
Part VI: Thinking about causes (extra if time)

Types of patterns: Differences between groups

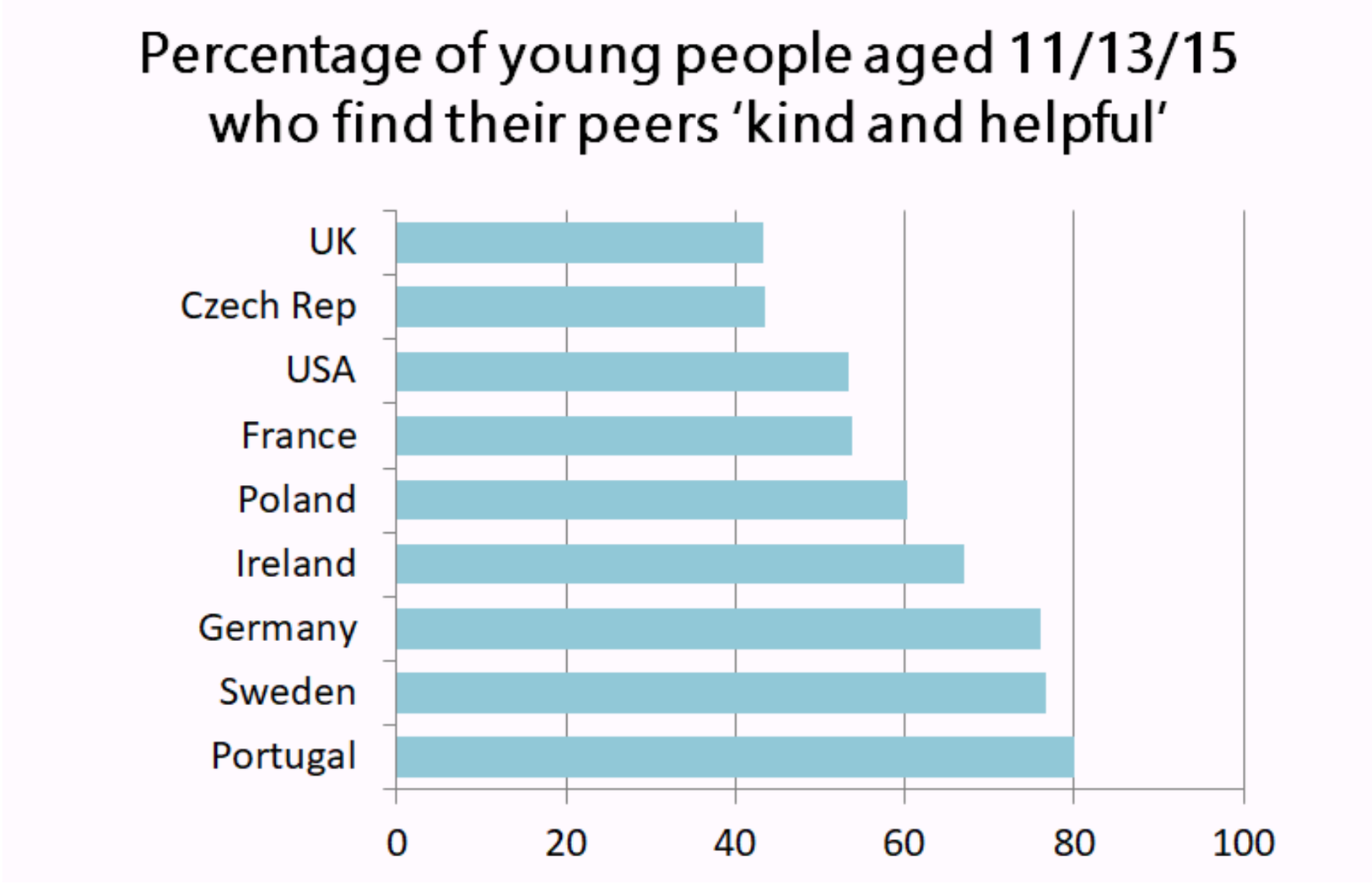


http://natcen.ac.uk/media/1319222/natcen_brexplanations-report-final-web2.pdf

Types of patterns: Social change



Types of patterns: Cross-national comparisons



Causal inference

Hypothetical causal relationship

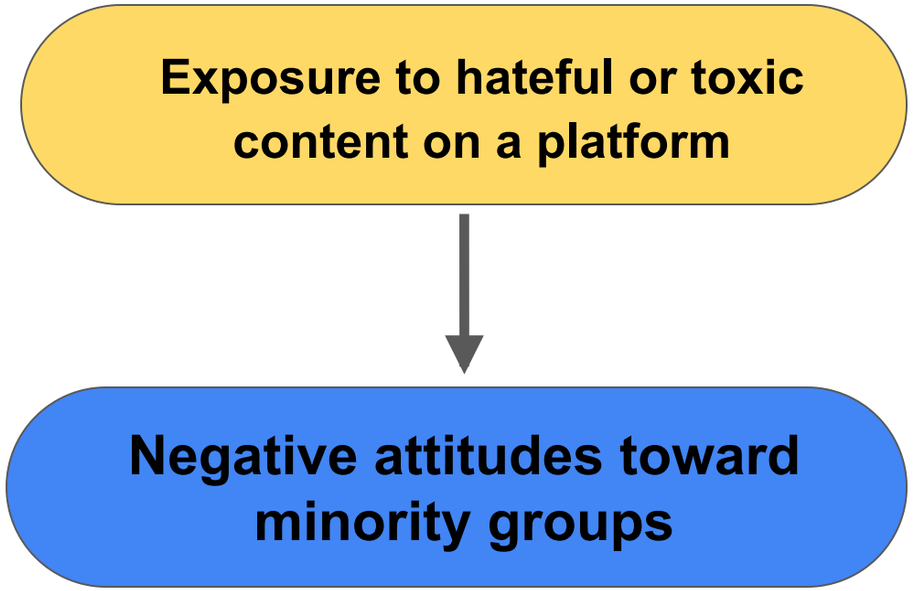
Exposure to hateful or toxic content on a platform

Negative attitudes toward minority groups



Quick note on language

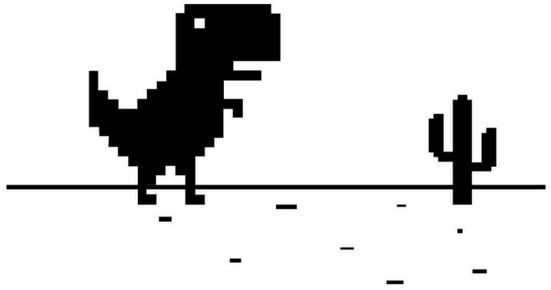
Hypothetical causal relationship



Independent variable (IV), causal, explanatory variable, predictor, etc.

Dependent variable (DV), outcome, symptom, response, etc.

Counterfactuals



!=



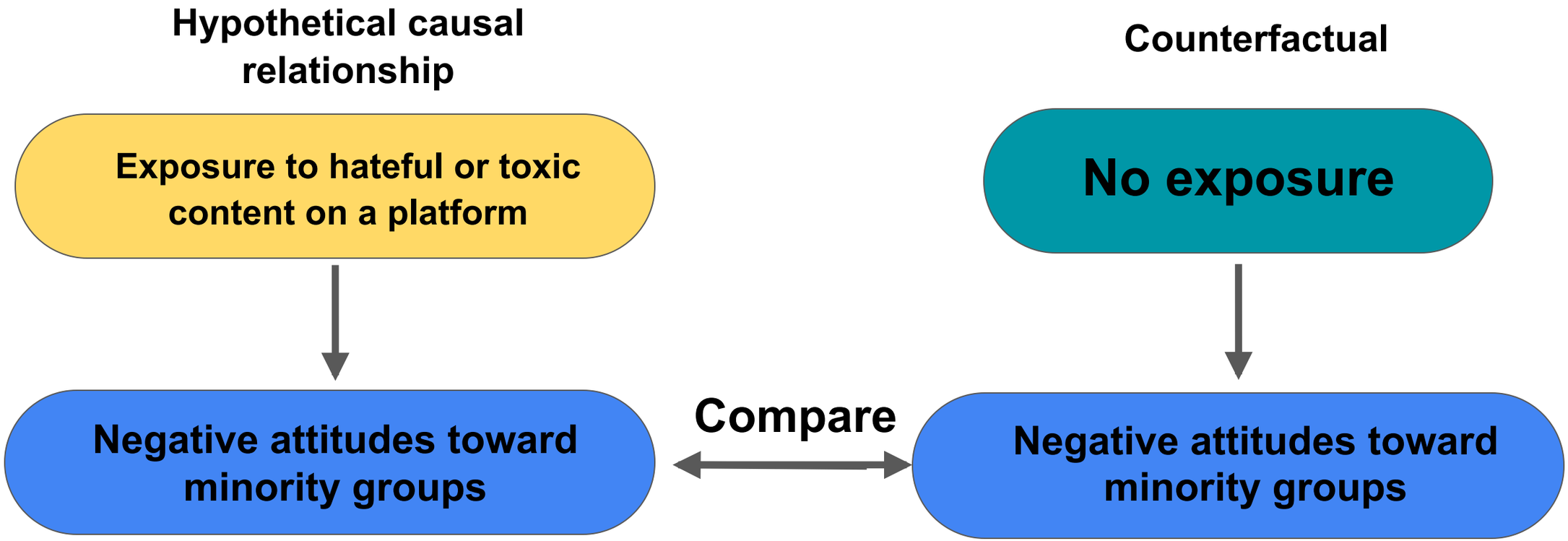
Counterfactual

Less exposure

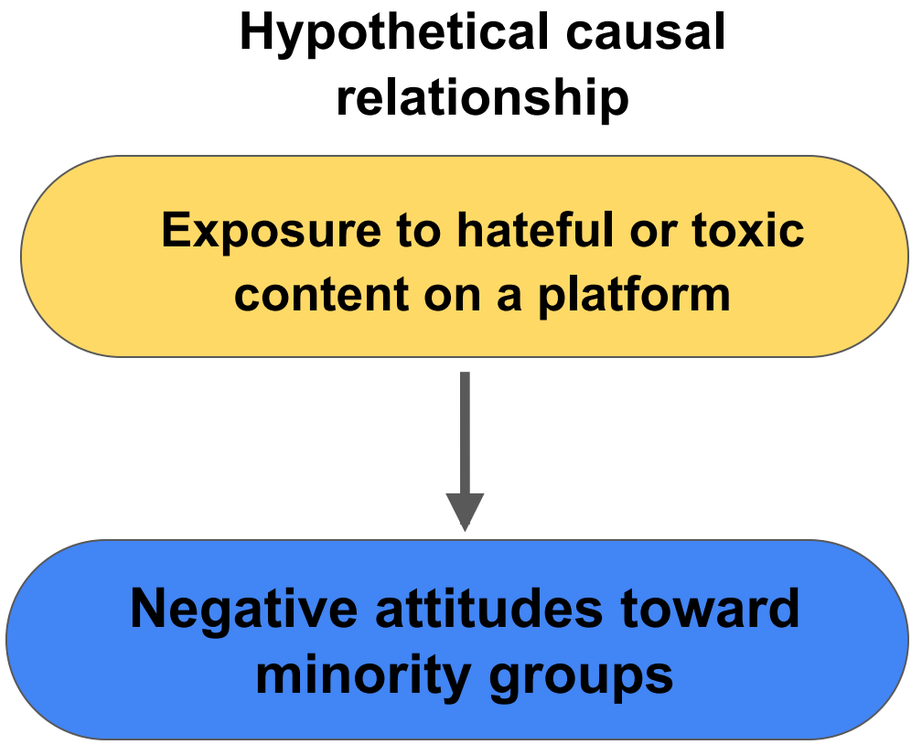
↓

Fewer far-right opinions

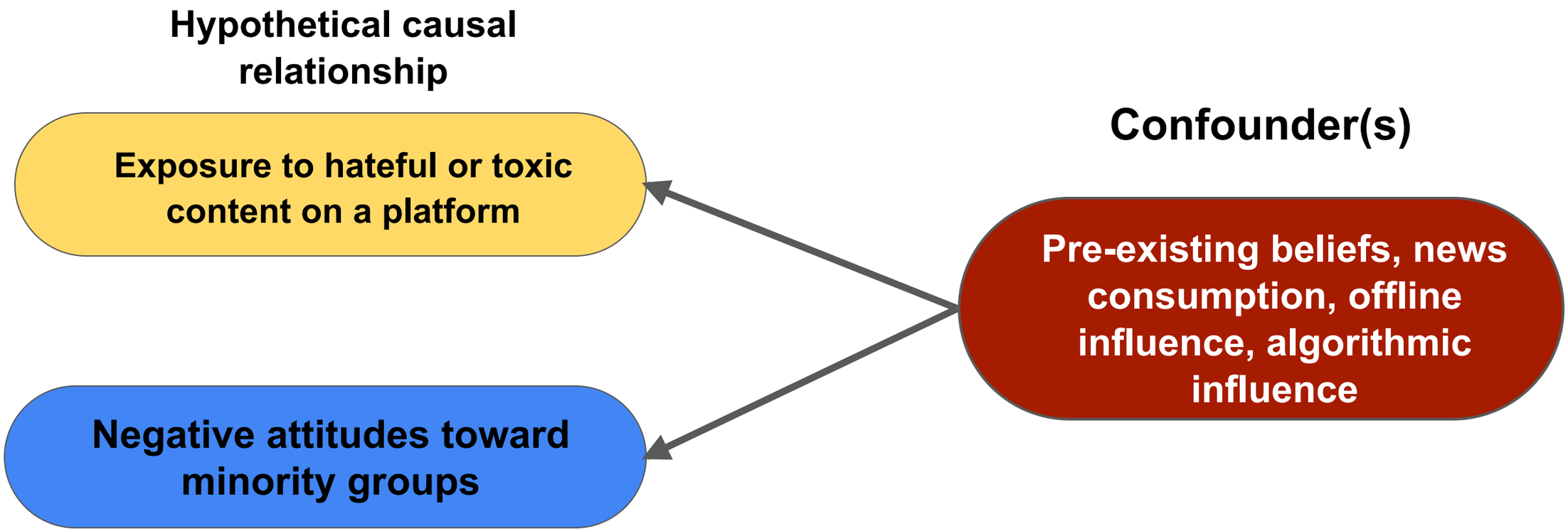
Causal inference and **counterfactuals**



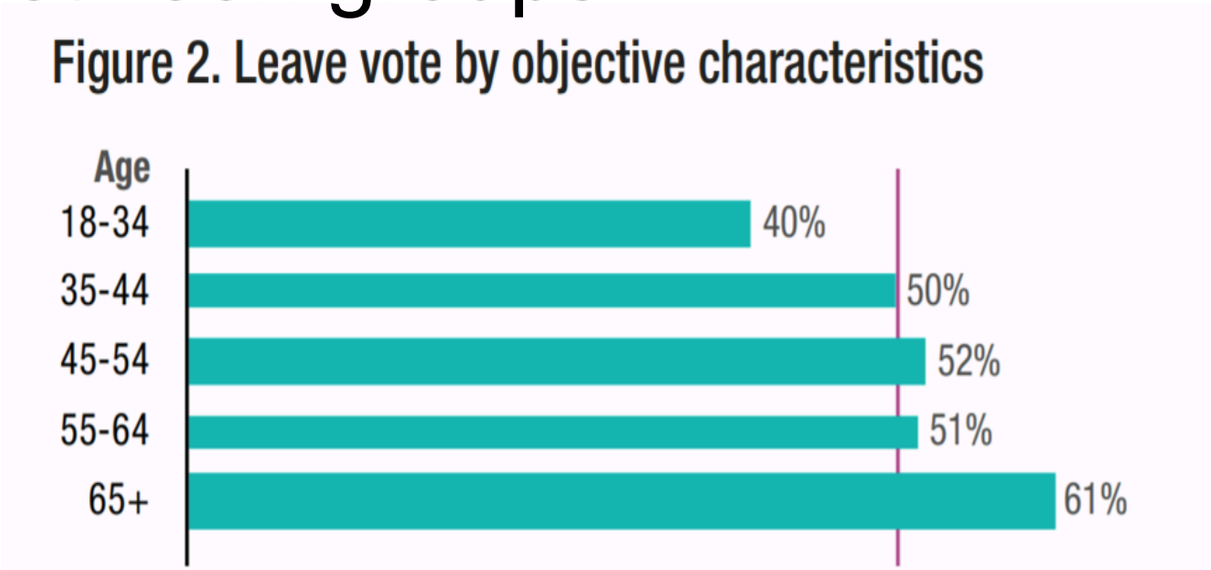
Confounding variables: a working counterfactual is not always enough



Confounding variables: a working counterfactual is not always enough

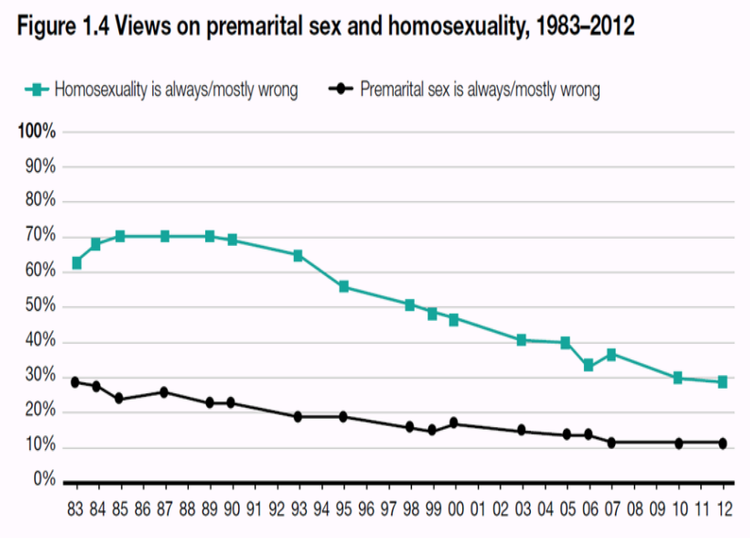


Types of patterns 1: Differences between groups

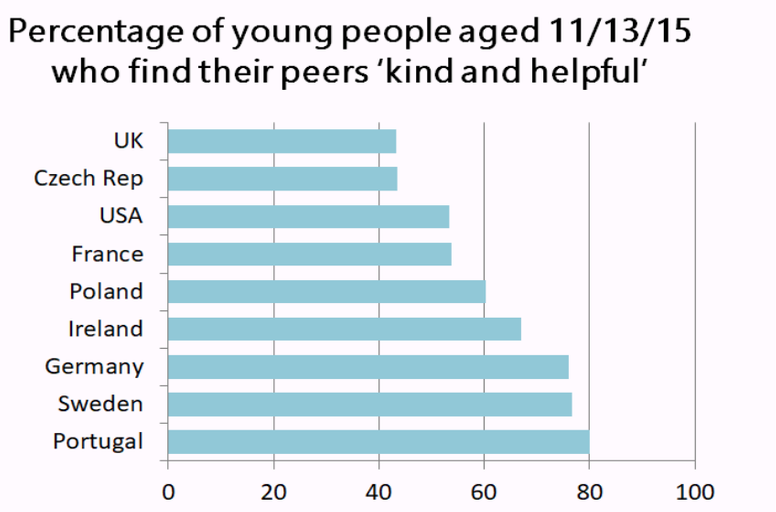


http://natcen.ac.uk/media/1319222/natcen_brexplanations-report-final-web2.pdf

Types of patterns 2: Social change



Types of patterns 3: Cross national comparisons



Quick note

This is the sort depth of analysis we want you to be thinking about.

We don’t need evidence of direct causal relationships (does A cause B) - this is often very hard to show. But it could be patterns between groups (are A and B different).

In either case counting is not enough - we don’t want an overview of a population or populations.

We want to understand how different variables relate to each other.

Part VII: Workshop

1. **Adapt the examples:** bars, histograms, lines and scatter plots.
2. **Explain the output:** what each observation and each axis represents.
3. **Save a figure:** labels, units and a caption that supports the interpretation.

Workshop: choosing and adapting plots

1. **Bar:** count venue records by borough.
2. **Histogram:** examine the distribution of borough venue rates.
3. **Line:** compare daily mean and median hire durations.
4. **Scatter:** plot two borough measures and calculate Pearson and Spearman.
5. Save and caption a figure. Continue with the optional extensions and your own data.

Before our first text lecture

- Complete this week's scatter-plot and correlation exercises.
- Read **Hill and Nulty, "Contextual Analytics"**, especially §§3–5.
- Bring one example of a representation choice that changes a possible claim.

Next lecture · 23 November: turn text into data, compare preprocessing choices, and introduce sentiment analysis.

We will keep asking: **what evidence does this representation preserve, and what else would we need to interpret it?**